

1. React :

A Javascript library used for building user interfaces, primarily for web applications. though it can also be used for mobile apps (via React Native)

2. React is a library not a framework

It handles only the view layer of an app (the V in MVC - Model, View, Controller)

You can combine it with other tools

React Router for routing, Redux for state.

3. Key Features

- * Component-Based Architecture

- * JSX (Javascript + ML)

- * Virtual DOM

- * Declarative UI

- * Unidirectional Data Flow

- * Hooks

- * State management

4. Single Page Application

A type of web application at website that loads a single HTML page and dynamically updates the content as the user interacts with the app without refreshing the entire page.

5. Component

A self-contained, reusable piece of the UI. Each component manages its own structure, logic, and appearance and can be reused throughout your app.

6. Functional Components.

In React, functional components are components defined as JS functions. They are one of the main ways to create components and also now the preferred approach because of their simplicity and ability to use Hooks for state and lifecycle features.

7. JSX

JSX - JavaScript XML. It's a syntax extension for JavaScript that allows you to write HTML-like code inside JavaScript. React uses JSX to define the structure of the UI in a more readable and declarative way.

8. Can browser understand JSX directly
No, browsers cannot understand
JSX directly.
JSX looks like HTML, but it's not
valid JS on its own

9. Babel
JavaScript compiler that converts
modern JS and JSX into plain
JavaScript that browsers can understand.

10. Virtual DOM
It's a lightweight, in-memory
copy of the real DOM that React
uses to improve performance and efficiency
when updating user interface

11. Difference between functional and class component
Functional Component : Simple modern and
uses Hooks

Class Components : Older more verbose
way using this.state
and lifecycle methods